

Zhongyang Li

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EDUCATION

The Ohio State University

PhD in Computer Science

Columbus, US

Aug. 2026 – Present

Johns Hopkins University

Master of Science in Computer Science | GPA:3.6

Baltimore, US

Sep. 2023 – May. 2025

Dalian University of Technology

Bachelor of Science in Software Engineering | GPA:3.8

Dalian, China

Sep. 2019 – June. 2023

RESEARCH INTERESTS

Large Language Models, Vision Language Models, Mixture-of-Experts, Test-Time Adaptation

PUBLICATIONS

[1] R2-T2: Re-Routing in Test-Time for Multimodal Mixture-of-Experts

ICML 2025

Proposed R2-T2, a test-time re-routing method for multimodal MoEs, achieving 5-10% accuracy gains on multiple benchmarks.

[Zhongyang Li](#), [Ziyue Li](#), [Tianyi Zhou](#)

[2] C3PO: Critical-Layer, Core-Expert, Collaborative Pathway Optimization in LLMs

COLM 2025

Developed C3PO, a test-time optimization method that improves MoE LLM accuracy by 7-15% through critical-layer and core-expert pathway optimization.

[Zhongyang Li](#), [Ziyue Li](#), [Tianyi Zhou](#)

[3] Routing Manifold Alignment Improves Generalization of MoE LLMs

ICLR 2026

Introduced RoMA, aligning routing-weight manifolds with task embeddings to improve MoE generalization via lightweight router finetuning, improving accuracy by 5-15% across multiple benchmarks.

[Zhongyang Li](#), [Ziyue Li](#), [Tianyi Zhou](#)

[4] Routing Entropy: A Hidden Self-Verifier for Free in Mixture-of-Experts LLMs

COLM 2026

Showed that the entropy of MoE routing weights serves as a free, training-free self-verification signal of output correctness, enabling reliable answer selection and confidence estimation at inference without any external verifier or reward model.

[Zhongyang Li](#), [Ziyue Li](#), [Tianyi Zhou](#)

[5] QUEST: Training Frontier Deep Research Agents with Fully Synthetic Tasks

arXiv 2026

Released QUEST, a family of open deep research agents (2B–35B) trained entirely on synthetic long-horizon search tasks via mid-training, SFT, and RL, achieving strong fact-seeking, citation grounding, and report synthesis across diverse benchmarks.

Jian Xie, Tianhe Lin, Zilu Wang, Yuting Ning, Yuekun Yao, Tianci Xue, Zhehao Zhang, [Zhongyang Li](#), ..., Yu Su, Huan Sun

RESEARCH EXPERIENCE

Research Intern - University of Maryland, College Park

Jun. 2024 - Current

Advised by Prof. Tianyi Zhou

College Park, US

- Conducted research on improving routing weights in Mixture-of-Experts (MoE) LLMs and LMMs through test-time optimization and router fine-tuning.
- Proposed and evaluated four novel methods: **R2-T2** (ICML 2025), **C3PO** (COLM 2025), **RoMA** (ICLR 2026), and **Routing Entropy** (COLM 2026).
- Demonstrated consistent 5–15% accuracy improvements across benchmarks and enabled smaller MoE models to rival or surpass larger dense models.

Research Intern - Johns Hopkins University

Sep. 2024 - Dec. 2024

Advised by Prof. Ashutosh Dutta

Baltimore, US

- Designed a hybrid watermark detection framework for LLMs by combining probabilistic token-group monitoring with Lomb–Scargle frequency analysis.
- Demonstrated high watermark detection success rates on models such as GPT and LLaMA while preserving text quality (under 1% BLEU score degradation).
- Enhanced the reliability of watermark-based attribution for LLMs while preserving natural text quality.

Research Intern - Dalian University of Technology

Feb. 2023 – Jun. 2023

Advised by Prof. Qiufen Xia

Dalian, China

- Proposed an asynchronous federated learning framework enhanced by blockchain and edge computing to address communication delays and trust issues in distributed training.
- Designed smart-contract-based mechanisms for secure model aggregation and incentive distribution, improving scalability and robustness of federated systems.
- Implemented a prototype system and validated its advantages in convergence speed and resilience to stragglers. Awarded DUT Outstanding Capstone Award (10%).

HONORS

US Mathematical Contest in Modeling Honorable Mention Award (15%)	2022
DUT First Class Scholarship (10%)	2023
COLM Travel Scholarship Award	2025

SERVICE

Reviewer: ICLR 2026, ICML 2026, NeurIPS 2026, COLM 2026, ICLR 2027
 Conference Volunteer: COLM 2025
 Teaching Assistant: Wireless Security (JHU)

CONFERENCE PRESENTATIONS

ICML 2025 (Poster)

R2-T2: Re-Router in Test-Time for Multimodal Mixture-of-Experts

COLM 2025 (Poster)

C3PO: Critical-Layer, Core-Expert, Collaborative Pathway Optimization for Test-Time Expert Re-Mixing

SKILLS

Language: TOEFL: 113 (R:30, L:30, S: 24, W: 29), GRE: 327 (V:157, Q:170, W:4.0)

Coding: Python/PyTorch, NumPy/Pandas, LaTeX, Bash, Linux, Git, Docker, Slurm

Machine Learning: Transformer, LLM pretraining/fine-tuning, vLLM, SGLang, DeepSpeed